

**RYK MODULAR
TIME SLICE
USER MANUAL
FIRMWARE V1.2**

Each track has Trigger Inputs/Output and CV Modulation inputs that can be routed to

a choice of destinations such as Record, Loop, Filter, Delay-time, Pitch Shift, Loop Splice Number, Restart, Reverse, etc etc.

MIDI CCs can be used as modulation sources, and MIDI NOTE data can be used as means to play loops as 1-shot samples polyphonically.

1. OVERVIEW

Time Slice has four Tracks each with a separate audio input and output. Audio is recorded to individual buffers, which can be played back and manipulated to their track outputs.

Additionally the Tracks can be mixed to the Stereo Mix output via the TRS MIX OUT jack.

Tracks can also receive audio from other track outputs or the Mix Output for resampling or recording a live performance based on the other tracks.

Each track has its own FX Chain that includes a low latency Pitch Shifter, Tilt Filter, Delay line, Wave Folder, Decimator and Reverb.

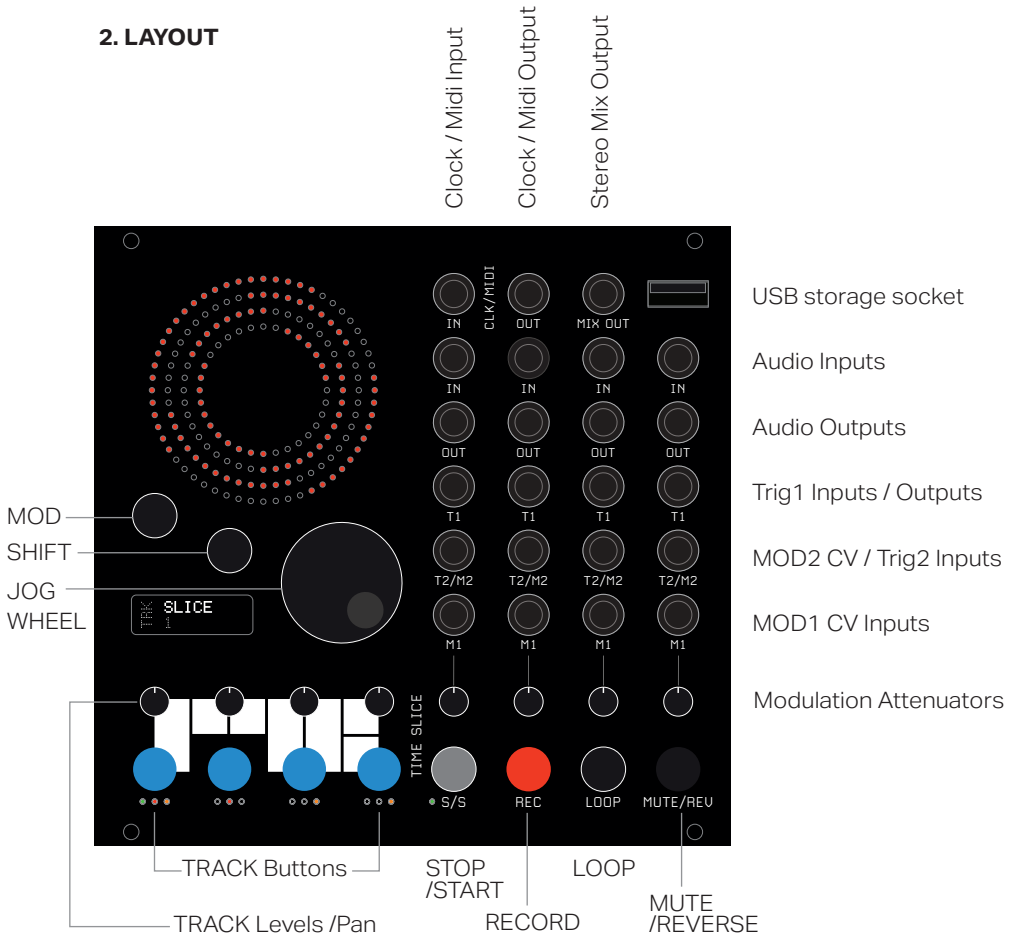
Time Slice can use Internal Clock sync (recommended), External Clock sync, or MIDI Clock sync.

The Loop and Delay units can be quantised time (based on the tempo) or unquantised time in milliseconds.

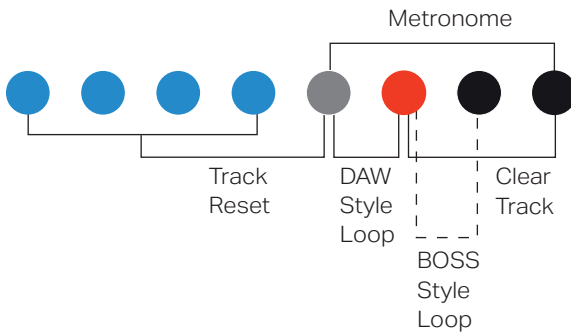
When using quantised Loop time, the LOOP and REC Button presses are quantised to the nearest 1/8th beat.



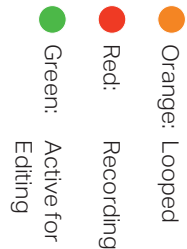
2. LAYOUT



2.1 BUTTON COMBOS



2.2 TRACK LEDS



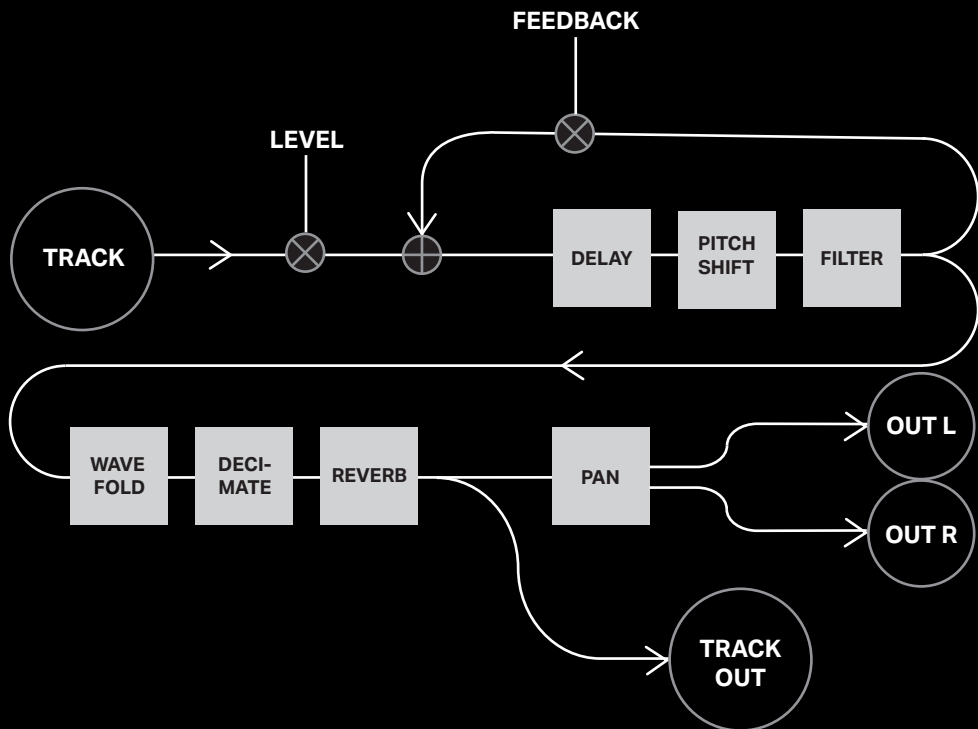


DIAGRAM: ARCHITECTURE

3. CONTROLS

- TRACK Buttons ●●●● – These are used to select a track for editing any settings, such as Loop Length, Modulation, Reverb, etc. The active track is indicated with a Green LED ● under the Track Button.

Two tracks can be Grouped to form a pair (eg. for stereo) by holding the SHIFT Button ●, and pressing two TRACK Buttons simultaneously.
Hold the SHIFT Button, ● and press a paired track to disable the pairing.

- S/S Button ● – Use this button to Start/Stop the internal clock. Loop-based functions require the clock to be running.

Alternative functions:

The S/S Button ● can be used to restart a Track playback by holding a TRACK Button ● then pressing the S/S Button ●. This is useful for retriggering 1-SHOT Loop mode samples

The S/S Button can also be used to toggle the the Metronome function On/OFF by holding the MUTE Button ● and pressing the S/S Button ●.

The S/S Button  is also used for starting a DAW Style Loop recording by holding the REC Button  and pressing the S/S Button . (See 4.LOOPING)

- REC Button  – Press the REC Button to engage recording for the active Track. A track that is recording is indicated with a Red LED  under the TRACK Button . If using BOSS style looping the first press of the REC Button  defines the start of the loop.

Alternative functions:

The REC Button  can also be used to clear the current track by holding the MUTE Button , and pressing the REC Button .

- LOOP Button  – Whilst the clock is running, the active Track can be looped by pressing the LOOP Button. The loop function is disengaged by pressing the button again, NB you will lose your audio loop when disengaging the loop. See the LOOPING section for more details on looping.

A track that is looped is indicated with an Orange LED  under the TRACK Button. If using quantised loop times, the start and end points are quantised to the nearest 1/8th beat.

- MUTE / REV Button  – Tracks can be muted by pressing the MUTE Button.

It can also be used as a Reverse Playback button by holding the SHIFT Button , then pressing the MUTE Button  to change the direction of playback for a looped track.

- LEVEL / PAN Trimmers (above the TRACK Buttons ) – These controls allow for adjusting the Output Level of a track, or Pan Position if using the Stereo Mix Output.

Turn the trimmer to adjust the output Level of the associated track.

Press and hold a TRACK Button  whilst turning the Trimmer to adjust the Pan position.

- SHIFT Button  

Press this button to toggle the menu between the Track Menu (Loop Time, Slice, Speed etc) and the FX Menu (Pitch, Filter, Delay, Reverb, etc)

Hold the SHIFT Button  and turn the JOG Wheel for finer adjustments of some parameters, such as Pitch in Cent values or fine Speed values, and alternate parameters such as Reverb Size etc.

Holding the SHIFT Button  and pressing the JOG Wheel will toggle certain functions such as Pitch Lock, and Speed Unlink (see individual sections below)

- MOD Button  

Press this button to toggle the menu between the Modulation Menu (CV, Trig, & MIDI etc) or the Settings Menu (Tempo, Inputs Sources, Clock, MIDI etc).

Press and hold the MOD Button  and turn the JOG Wheel to adjust additional parameters such as Delay Mix, Reverb Mix etc (see individual sections below)

- JOG Wheel – The JOG Wheel is used to edit all the settings within the menu system. When the top-level menu is displayed in bold, turning the wheel scrolls through the menu items.

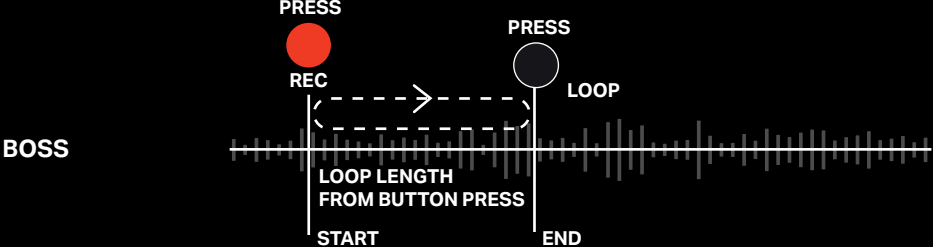
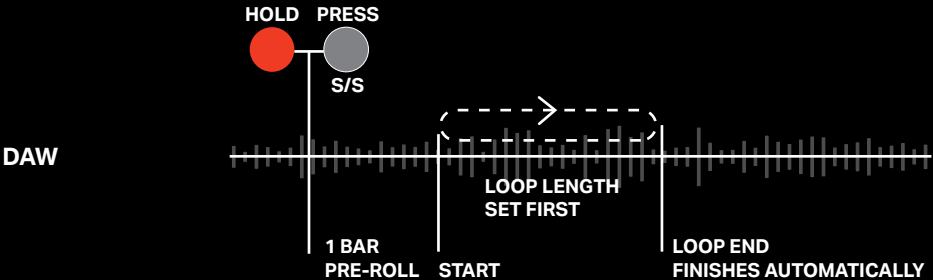
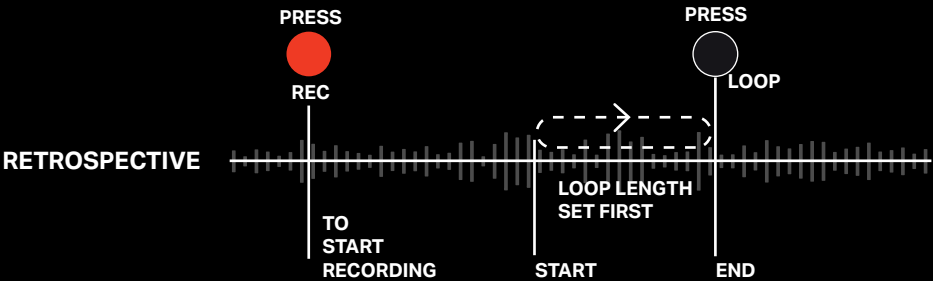
Press the JOG Wheel to toggle between

the menu item, and it's parameter value.
With the parameter value displayed in bold,
turning the JOG Wheel adjusts the value.
Some alternate and fine adjustment of
parameters can be edited by holding the
SHIFT or MOD Buttons.(see individual
sections below)

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NB. Please note the JOG Wheel has a
unique soft touch press switch, be aware
that it doesn't require much force to
activate, and should be pressed in the
centre NOT at the edge.

THREE LOOP STYLES



- Modulation Attenuators (Below the M1 inputs) – These controls adjust the attenuation of the M1 CV inputs.

If there is no jack inserted into an M1 Input, it can be used as modulation control. (see MOD1 Modulation menu)

4. LOOPING

To use the Looping functions, the internal or external clock needs to be running, this is indicated by a flashing Green LED ● next to the S/S button.

Press the S/S Button ● to start the internal clock (recommended) , or use a running external Clock.

4.1 LOOP STYLES

There are three ways of enabling Loops on Time Slice, RETROSPECTIVE, BOSS and DAW style. (see the diagram on page 5) The Loop Styles RETRO, or BOSS can be chosen in the Settings menu. (see Settings Menu)

- RETROSPECTIVE

The endpoint of the loop is defined by when the LOOP Button ● is pressed, and the start point is the moment back in time previously defined by the Loop Length.

In the Track menu (TRK) navigate to the LOOP UNIT item, and adjust the setting to QUANTISED, to use tempo based time divisions.

Then navigate to LOOP and adjust the length of the Loop, say to 2 bars. Now press the REC Button to start recording, and perform some music or sound, When you are happy with the last two bars, press the LOOP Button ● to engage the Loop.

NB when using quantised Loop units, the LOOP Button activations are always quantised to the nearest 1/8 beat, so you will need to think about the timings of when to set the loop.

The Retrospective Loop style allows the user to perform and record repeated phrases continuously, and when happy with a phrase, Loop can be engaged to capture the last phrase played.

Try engaging the Metronome (hold the MUTE Button ●, and press the S/S Button ●) to get a feel for the Bar divisions for beat based loops when performing.

The Metronome plays every 1/4 beat, so 1 bar is represented by 4 Metronome blips.

For example if making a 1 bar loop, hit the LOOP Button on the 5th Metronome blip after playing 1 bar of music or audio.

- BOSS

With BOSS Loop style the Loop length is defined by the time between pressing the REC Button ●, and then the LOOP Button ●.

To make a BOSS style loop, change the LOOP STYLE menu setting to BOSS, make sure the clock is running.

Press the REC Button ●, perform some sound, then when finished press the LOOP button ● to create the loop.

Again, if using quantised loop units, these button presses are quantised to the nearest 1/8 beat

- DAW

With DAW Loop style, the Loop length is set first, and then a timed recording is made for the duration of the Loop.

First stop the clock by pressing the S/S Button ●.

In the Track menu navigate to LOOP, and set the desired Loop length.

To start recording press and hold the REC Button ●, then press once the S/S Button ●.

The clock will preroll with 4 beats of Metronome blips, then start recording for the duration of the Loop length.

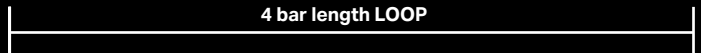
Once the length of the loop has been recorded the Loop will automatically set and recording will turn off.



DIAGRAM : EXAMPLE OF USING LOOP LENGTH BASED SLICES

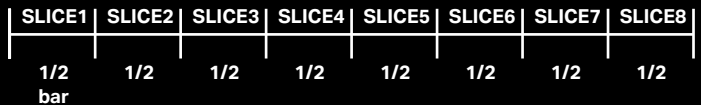
RECORD A 4 BAR LENGTH LOOP

TRK LOOP
4 Bars



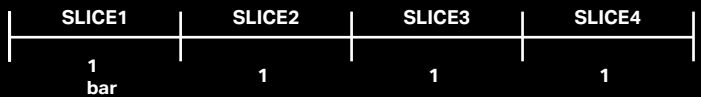
THEN CHANGE LOOP LENGTH TO 1/2 BAR TO CREATE 8 SLICES

TRK LOOP
1/2



OR LOOP LENGTH TO 1 BAR CREATING 4 SLICES

TRK LOOP
1 BAR



THEN CHOOSE WHICH SLICE TO PLAY

TRK SLICE
2



A Slice is a shorter piece of a recorded Loop or Loaded audio.

Slices are normally loop length based, but alternatively that can be based on markers placed in the audio. (see SLICE MODE in Track Menu below)

Once a length of audio has been recorded or Loaded, in the Track menu adjust the LOOP length to a shorter length than the original, to play smaller sections of the audio, called SLICES.

The menu parameter SLICE is the Slice number that you will hear when a track is playing the new shorter loop.

Using Loop Length based slices for example, record a 4 Bar length Loop then adjust the Loop Length to 1/2 Bar, the Slice range will be 1-8, allowing you to select 1 of 8 Slices from the original loop. Slice number 1 is the first slice of the looped audio, and represents the beginning of the original loop.

Selecting negative Loop slices -1, -2, -3, etc., will jump back in time.

This is great if you want to play several variations of the same phrase and after setting the Loop, you can go back in time and jump to these different versions of the phrases.

the technique of Granular synthesis.

Try this with a piece of longer looped audio (2 bars or more) then set the Loop length to a short value, such as 1/32 or 100ms, then scroll through the SLICE numbers to hear this Granular scanning technique.

You can also use a CV to scan through the buffer. (see EXAMPLE USES FOR MODULATION INPUTS)

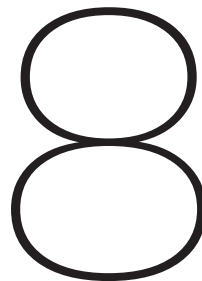
If you have a tempo synced break beat loop of 2 bars or so, you can set your loop length to something shorter like 1/4 or even 1/8 bar, and use the SLICE number to jump around to different parts in the break beat, much like the way drum-and-bass music is made.

Try LOADING the Track

"break_beats_100bpm.wav" from the internal memory and adjusting the Loop length to 1/4 bar, then adjust the SLICE number to hear the individual slices playback.

Slices can also be modulated with CV or MIDI, for highly creative restructuring of audio loops. (see Modulation Menu)

Or you can go micro, and scan through the looped audio in very small loop sizes, like



4.3 LOOP MODES

In addition to the normal Loop Mode, there are two other types of Loop Modes; 1-SHOT and LIVE.

In the Track menu, navigate to LOOP MODE, to select.

- 1-SHOT

This mode acts more like a traditional sampler, where the Loop plays just once then stops. To use this mode either LOAD a WAV sample or record a loop in the normal way.

Playback is triggered by a RESET trigger to a TRIG input, or MIDI CC, or MIDI NOTE (see Modulation Menu)

You can also manually trigger playback by holding the TRACK Button ●, and pressing the S/S Button ●.

The XFADE value acts as a decay amount fading the level of the playback towards the end of the sample. [0 – no fade, 50% - the sample fades from halfway to the end]

1-SHOT normally plays to the end of the loop, but this can be changed to GATED mode where the sample stops playing when the gate level turns off. (see 1-SHOT in Settings Menu)

- 1-SHOT POLY

This mode is the same as 1-SHOT but enables four individual voices of polyphonic playback, this can be great to use with a MIDI keyboard, or for creating complex layers of overlapping samples.

- LIVE

This mode creates a looping buffer behind the record head of the main buffer, much like a live single-grain granular effect. With the clock running, press the LOOP

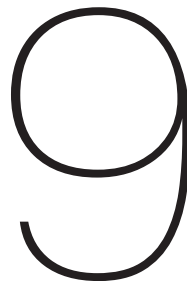
Button to engage looping, then activate Record by pressing the TRACK Button and REC Button.

The contents of the loop are only updated by a SLICE value change, or RESET trigger via a TRIG input, or MIDI CC etc.

You can also manually trigger the loop update by holding the TRACK Button ●, and pressing the S/S Button ●.

It is a great tool to create glitch or stutter effects from live audio input.

(See EXAMPLE USES FOR MODULATION INPUTS)



5. MENU

5.1 TRACK MENU



The Track Menu is where all the Loop orientated settings of a track can be adjusted. The settings apply to the currently active track, indicated by the the Green LED ● under the TRACK Button.

- SLICE MODE – With this setting SLICES can be changed from the default mode based on Loop length multiples, to MARKERS which are manually or automatically placed slice points (see SLICE EDIT below)

- LOOP MODE – Here you can choose the loop playback mode, from **NORMAL**, **1-SHOT**, **1-SHOT POLY**, or **LIVE** mode.

- LOOP UNIT – Choose between **QUANTISED** for beat/bar based loop units, or **NON QUANT** for millisecond units.

- START – This is the Start position of a looped track buffer, intended for fine-tuning loop start points such as drum breaks or when using 1-SHOT sample playback, but can also be used for scanning through a longer piece of audio whilst using a short Loop length.

When the clock is stopped, use the JOG Wheel to scrub the START value of the looped audio to find start positions of beats and notes etc.

- LOOP – This is the length of the loop..Before making a **RETRO** or **DAW** style loop,

this should be set first.

- SLICE – Indicates which Slice will be played back from the buffer when there is an active loop.

Slice #1 is the start point or first slice in the Loop.

Positive numbers are forward in time, using shorter than original loop-lengths allows for scanning through the looped audio.

Negative numbers go back in time to before the loop was created. (see LOOP-ING above)

If using Slice Marker mode, the slice numbers are limited to the amount of markers that are defined.

These can be created in the SLICE EDIT menu, or are loaded with a WAV file or track file that contains markers.

- SLICE EDIT – This menu allows for the creation of MARKERS for slices instead of loop length based slices.

It has 3 sub menus, which can be accessed by holding the MOD button ● and turning the JOG Wheel.

- AUTOSLICE – Autoslice will go through the loop and place Markers at strong transient points such as drum hits or note onsets. This is great for creating individual slices out of breakbeats or making multi-samples from an array of sounds such as the notes on a piano. (See examples section)

Long press the JOG Wheel to start slicing !

- ADD – Long press the JOG Wheel to add a marker point at the current position in the loop. When the clock/playback is stopped, the JOG Wheel can be used to scrub the audio to locate a position to add a marker. Adding marker points can also be done

whilst playing, though it is easier if you set the playback speed to 50% or less.

- DEL – Use the JOG Wheel to select a marker, then long press the JOG Wheel to delete the selected marker.

- SPEED This is the playback speed of a Looped track. The values are quantised to 25%, 50%, 100%, 200%, 400% for quick access to useful speed changes.

For non-quantised speed values, right down to 0% , hold the SHIFT Button ● while adjusting the JOG Wheel.

NB for performance friendly speed changes, the quantised values can be restricted to updated only on bar boundaries using a setting called Parameter Sync (see PARAMETER SYNC in the Settings Menu)

The speed of a loop can be “Unlinked”, so that it is uninfected by the Tempo of the clock. This is useful for pitched 1-Shot samples such as piano, where you want to keep the samples at the correct pitch without them being affected by the changes in Clock tempo.

Hold the SHIFT Button and press the JOG Wheel to UNLINK the Speed.

- XFADE – This value controls the amount of cross-fading from the end of the loop to the beginning. The value is based on the percentage of the total loop time.

This is useful for creating softer looping. NB Please note that XFADE uses audio ahead of the loop start point, so unexpected results may occur if unwanted audio exists before the loop start point.

When using 1-SHOT loop mode, the X-fade controls the length of the fadeout of end of the sample.

- SOS MIX – Once a Loop is created or a sample is Loaded it can be overdubbed with new audio material, this is called Sound on Sound (SOS).

The SOS Mix amount can be adjusted from 0% - New audio only, to 100% New + Looped audio

- L-TEMPO – This value represents the tempo of the loop when it was created using the LOOP Button, or the stored Tempo of a Track that is has been Loaded. It can be adjusted after creating a loop to fine tune the loop timing, which can be useful for getting a break-beat to loop more precisely.

When recording a drum track or beat based music as an input for a loop it is usually best to set the TEMPO BPM to match the source music. (see Settings Menu)

You can S/S the clock, and use the internal Metronome to get a good idea if the Time Slice tempo matches the audio, before you proceed to record and loop.

- LOAD / SAVE –You can Load and Save audio Buffers, Tracks, and Projects to and from the internal memory or to a USB Flash drive. (See STORAGE in the Settings Menu) Hold the SHIFT Button ● and turn the JOG Wheel to toggle between Buffer, Track, or Project for Loading or Saving. Select the file name, and press and hold the JOG Wheel for 1 second to activate a LOAD or SAVE.

NB Loading and Saving of files can take some time, especially stereo files, whilst Loading and Saving the LED display will freeze. This is normal behaviour.

You can load files in any sample rate, at 16,24, or 32 bits and Time Slice will

resample the audio using a high quality algorithm to its native format of 16bit 48khz.

If a non grouped track is selected only the Left channel of a Stereo file is loaded.

If loading a Stereo file, and a grouped track is selected, the Left channel is loaded to the left most track, and the Right channel is loaded to the right most track.

If loading a Mono file to a grouped track, each track is loaded with the same mono data from the file.

WAV files that include Marker data can be imported, this enables the SLICE Marker option which uses the markers as slice points in the loop.

If using DAW software such as Reaper to export markers, please select the option "markers only" on export.

Directory browsing is supported for Loading and Saving of files.

Audio files Saved from Time Slice are files named "sample_A.wav", "sample_B.wav" and so on.

Project files are named "project_A.tsprj", "project_B.tsprj" etc.

Files on a USB drive can be renamed at a later date, but the files have to keep their extension suffix as the ".wav" or ".tsprj".

5.2 FX MENU



Here you can edit the FX chain of Effects that are applied to each track.

P-SHIFT – The PITCH SHIFT effect can be used to create harmonies, re-tune samples, or create time-stretch style effects. It can be tuned up 12 semitones, or down 24 semitones.

Hold the SHIFT Button ● to fine-adjust the pitch in cents.

Hold the MOD Button to adjust the dry/wet MIX of the Pitch Shift. (NB only when DELAY time is 0)

Hold the SHIFT Button and press the JOG Wheel to toggle the PITCH LOCK mode. PITCH LOCK retains the pitch of the track even when the Tempo or Speed of the Loop are adjusted, this can be used to create a time-stretch effect.

For a classic harmonised type sounds on instruments, set the pitch to musical intervals such as -3, -5, -7 etc and set the Mix value to 50%

For a chorus type effect set the pitch to a small value such as -0.100 and set the Mix value to 50%;

Using a Pitch Shift of 1 or 2 semitones with a short DELAY effect, can create amazing downwards spiralling effects on drum tracks or short percussive sounds.

- DELAY – This is the delay time of the delay effect, a value of "0" turns off the delay effect.

Hold the SHIFT Button ● to adjust the Delay FEEDBACK amount.

Hold the MOD Button ● to adjust the dry/wet MIX of the Delay.

Use with the PITCH SHIFT effect and set

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musical intervals such as +2 or -3 and longer delay times and 50% Feedback, to create ascending or descending scales from single note sounds.

- DELAY UNIT – Choose between QUANTISED for beat/bar based delay times, or NON QUANT for millisecond times.

- REVERB – This is the send level to the REVERB effect.
Hold the SHIFT Button to adjust the Size of the reverb effect, values below 20 create the effect of a smaller room space, values above 60 create the effect of large hall like spaces.
Press and hold the MOD Button to adjust the dry/wet MIX of the Reverb.

- FILTER – This is the cutoff frequency of the HP/LP tilt filter.
This is a classic DJ style tilt filter, where adjusting the cutoff frequency passes the mode from Low Pass to High Pass mode.
Hold the SHIFT Button and dial the JOG Wheel to adjust the resonance amount. (NB high settings of resonance may create saturation-distortion when used with the DELAY effect)

To reduce muddiness in DELAY or REVERB effects, try setting the FILTER HP frequency to a low value such as 100-200, eliminating some of the lower frequencies for a cleaner effect.

For increasingly warmer DELAY effect repeats, try setting to a LP frequency value such as 2khz to reduce the high frequencies in the delay feedback path.

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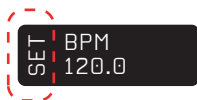
- WAVE FOLD – This applies a Wave Fold / Saturation type effect to the audio. Larger values create an unusual distortion effect. Sounds good on bass drums, and simple pitched sounds.

- DECIMATE – This is a digital degrade effect that reduces the audio quality using a combination of interpolation styles, bit depth reduction, and sample rate reduction.

- AMP – Here you can add additional gain to the signal, which can be useful for low level sounds.

- M-COMP – This is a master compressor that is applied to the Stereo Mix output only.
Use the JOG wheel to adjust the ratio such as 1:4, hold the MOD button ● and use the JOG Wheel to adjust the Threshold, hold the SHIFT button to allowing adjusting the Makeup gain.
A good starting point for settings is:
Threshold – 50%, Ratio – 1:4, Makeup 75%.

5.3 SETTINGS MENU



This menu is used for adjusting the Global settings of the Time Slice module, as well as some Track based settings such as Input Source etc,

- BPM – When using the internal Clock, this allows the tempo to be adjusted. When using an external clock source, the detected tempo is displayed here.

- TAP BPM – When using the internal Clock, the tempo can be set by tapping the MOD Button to a consistent 1/4 beat for 4 beats or more.

If recording from a beat based source such as a drum break, it can be useful to use the Tap Tempo to get the tempo matched before recording. Set the Metronome on (press Mute and S/S) and start tapping, using the S/S button to restart the clock to see if the Metronome blips match the beat of the music.

- INPUT – Here you can choose the audio input Level or Source, for the currently selected track.

For the Audio Input jack, the level can be chosen from LINE Level, EURO-rack level, or HOT EURO level. (HOT EURO is for signals that are greater than a +/- 6V range)

This value is per track, the track number is indicated in the top right corner.

Please make sure to select the correct level to avoid a CLIP ! overload of the signal. NB the Track Inputs are normalised from Left to Right, so be aware that a track that is not set to the correct level may still clip

even without a jack in it's input.

Alternatively the audio input can be taken from the output of another track, or the Mix Output, enabling resampling of tracks.

This is similar to the four-track tape technique known as track bouncing, where existing tracks are mixed and recorded to another track.

- LOOPSTYL – Select the Loop Style from RETROspective or BOSS for loop recording.

- RSET TRK – Use this option to reset the currently selected track parameters to the default settings. This resets all the track, modulation and FX parameters. Select "YES" and press the JOG Wheel for a 1sec.

You can also select RESET ALL, to reset all the tracks !!

- INPT MON – The Input Monitor option toggles On/Off the preview of the audio input, which is mixed to the currently selected track.

- METRNOME - The Metronome can be turned ON/OFF and it's output can be set to ACTIVE TRACK / ALL TRACKS / OUT1 / OUT2 / OUT3 / OUT4.

NB the Metronome can also be turned On or Off by holding the MUTE button and pressing the S/S button.

-1-SHOT – This allows setting the gate mode of the 1-SHOT loop mode. NORMAL plays the whole loop when triggered, GATED plays the loop until the gate signal goes low.

NB this value is per track, the track number is indicated in the top right corner

- PRAMSNC – Parameter Sync quantises changes of the SPEED parameter to bar boundaries.

For example if you want to drop the speed of a break beat to 50%, and later back up to 100%, setting Parameter Sync ON will restrict the speed changes to only occur on the 1st beat of bar retaining the timing of the beat.

- STORAGE – Here you can choose where to Save and Load Tracks and Projects. It can be either to the Internal memory or to a USB flash drive if you have one. NB when inserting a USB flash drive the display will freeze for an instant whilst the device is recognised, please do not pull out the USB drive whilst this is happening. Do not pull out the USB drive whilst Loading or Saving.

- CLCK SRC – The Clock Input can be set to Internal, External Clock, or MIDI Clock. Internal Clock is recommended for the best audio fidelity when overdubbing loops.

- CLCK OUT – The Clock Output can be set to analogue 4PPQN, 24PPQN, or MIDI Clock.

- CLCK DIV – The Clock Input division when using an External Clock source. It is recommended to use a value of 4 for use with other RYK products. This equates to 4 pulses per quarter note (PPQN) which is the equivalent to 16th beats.

- MIDI CH – The MIDI Channel number for MIDI input of MIDI CC or MIDI note data. This value is per track, the track number is indicated in the top right corner.

- SM LTNCY – When using RESET Trigger and SLICE modulation from CV, some devices such as CV/GATE keyboards can have some latency between the Gate signal and the CV signal.

To alleviate this, set the LATENCY value to HIGH.

- MENU SYS – There are two settings for the menu system, LOCAL and GLOBAL. For beginners we recommend using GLOBAL, this uses the same menu position for all tracks, making the menus system easier to understand. LOCAL keeps the menu position between tracks, allowing different values to be edited for each track more quickly, but this can be confusing on first use of the module.

- MENU TIP – Menu tip pop ups show up on certain menu items such as REVERB, and indicate briefly the alt functions available with the MOD and SHIFT buttons. For beginners we recommend having MENU TIPS switched ON.

- FIRMWARE – From time to time we release firmware updates. To install a new update. Copy the firmware file to the root of a USB stick. Insert the stick into the USB socket on the front of the module, and power up normally. Navigate to the FIRMWARE menu option, and select update. Long push the JOG Wheel to activate the update. The display should read UPDATING,

DONE them finally REBOOT if successful.
Reboot the module when finished.

5.4 MODULATION MENU



This Modulation Menu is where you can assign modulation CVs from the M1,M2 inputs, & MIDI CCs to destinations such as Speed, Pitch Shift, Slice, Feedback etc. Also the T1, T2 trigger inputs and MIDI CCs can be assigned to events such as Reset, Record, Reverse, Mute etc. The T1 jacks can also be used as a trigger outputs for End of Loop Pulses, and Transient detect pulses.

The Clock IN jack can be used as a MIDI INPUT for modulation if not using External Clock.

Press the MOD Button to access the Modulation Menu, use the JOG Wheel to select a Modulation Source such as MOD1 (the top menu item in bold). Then press the JOG Wheel to toggle to the parameter edit, and select a modulation destination. NB The destination will ONLY become active when you press the JOG Wheel again to escape to the top level of the Modulation Menu. (MOD1, MOD2 etc in bold)

MOD1, MOD2 – Here you can route a CV from the M1 or M2 inputs to a destination parameter for the selected track. The trimmer below the M1 socket is an attenuator for the M1 CV. If MOD1 is selected, and no Jack is inserted into the M1 socket, the attenuator acts as a control knob for this parameter. This can be useful for setting up manual controls for certain parameters.

Some modulation destinations are marked "Q", such as P-SHIFT_Q, SPEED_Q, SLICE_Q. The "Q" option indicates the CV values are quantised to V/OCT semitones. So for example a CV used to modulate SPEED_Q will always quantise to the nearest semitone.

TRIG1 IN, TRIG2 – Here you can route a trigger/gate signal from the T1 or T2 inputs to a destination event. Destinations options that are marked "TOG" such as REC_TOG indicate that a low-to-high pulse will toggle the function On, and another low-to-high pulse will toggle the function Off. Other destinations such as MUTE are continuous, and switch On when the trig input goes high, and switch Off when the trig input goes low.

Destination options that are marked with "->" denote that trigger can be Time Quantised. Hold the MOD button and adjust the JOG Wheel to set the quantise value.

Three LOOP options can be used to activate the three LOOP styles, the first trigger will start the Recording of RETRO and BOSS, the second trigger will set the Loop.

TRIG1 OUT – The T1 jacks can also be used as trigger outputs for EOP (end-of-loop pulse), and TRANS transient detection pulse.

The TRANS80 and TRANS50 options send a brief pulse out when the audio level of the track exceeds 80% or 50% of it's average value.

EOP will send a brief pulse out each time the track's loop reaches it's end and restarts.

The EOP pulse can also be clock multiplied or divided, creating multiple pulses per loop like a clock, or periodically every 2,3rd, 4th etc loop.

Hold the MOD button and adjust the division or multiplication ratio with the JOG Wheel.

MIDI NOTE – This allows the use of MIDI NOTES to trigger the Reset of the Loop for 1-SHOT style sample playback, and use the NOTE value to control the pitch of the loop via Speed, or Pitch Shift .

Alternatively the NOTE value can be used to select Slices within the loop. This can be used for playing multi-samples within a WAV file, where each note or hit is aligned either with a marker in the file, or aligned to BPM time divisions such as 1/2 bar intervals.

(see Examples below)

MIDI CC a-d – There are four assignable modulation slots for MIDI CC's for each track.

Hold the SHIFT Button, and press the JOG Wheel to enable MIDI Learn for the selected MIDI CC modulation slot. The number of the CC will flash on the top right corner whilst the MIDI Learn is active. You can also choose the CC number by holding the SHIFT Button and dialling in the JOG Wheel.



EXAMPLES USES FOR MODULATION INPUTS

Granular Scanning

in the Track Menu, set a loop length to 1 Bar or longer. Record some audio and press the LOOP Button to make a loop. Now set the Loop length to a small fraction such as 1/32.

In the Modulation menu, set the MOD1 destination parameter to SLICE, and remember to toggle back to the MOD1 option to make the setting active. Use the M1 Trimmer or apply a CV to scan through the audio buffer creating a granular style effect. Adjust the Xfade value to soften the windowing of the grain.

Pitch Shifted Sample Player

Record a section of audio, and press the LOOP Button to make a loop. Adjust the Start parameter so that the playback starts at a preferred point (you can preview this by using the S/S Button to start and stop the playback)

In the Track Menu change the LOOP MODE to 1-SHOT mode.

In the FX Menu set P-SHIFT to -24, and make sure Pitch Lock is not active.

In the Modulation menu, set M1 to modulate P-SHIFT_Q, and set T1 to the RST

destination.

Connect a keyboard or sequencer CV to the M1 input (with the attenuator to max) and connect the Gate signal to the T1 jack. When notes are played the keyboard or sequencer, the sample will be retriggered and it's pitch controlled by the CV. NB the lowest note will be 0V, highest note 3V.

Add an envelope to the setup. Patch the envelope output to M2, and set the MOD2 destination to Level, then turn the level Trimmer to minimum. Connect the envelope gate to your keyboard or sequencer gate.

For a more traditional approach change the MOD1 destination to SPEED_Q instead of Pitch. In the FX Menu set P-SHIFT to 0. In the Track Menu set Speed to 25%. Now when you play the keyboard or sequencer, the speed AND pitch of sound changes.

Multi-Sampled Drums

In the Track menu, find LOAD and toggle to TRK for track loading a WAV file (hold the SHIFT Button and turn the JOG Wheel). Turn the JOG wheel to locate the file "drums_1shot.wav" from the "1-shot" folder in internal memory. Long press the JOG to load the file.

In the modulation menu, find TRIG1 and select the RST option, and remember to toggle back to the TRIG1 option to make the setting active. Then find MOD1 and select SLICE_Q, again toggle the menu back to MOD1 to activate the setting. Connect a sequencer CV to the M1 input for the track, and the Gate output to the T1 input for the track. Use the pitch values of you're sequencer to

select the different SLICES to play back different drum samples, the Gate will trigger when the drum samples play. (NB make sure the clock is running, otherwise the loops will not play)

Using AUTOSLICE to create a multi-sample instrument

Set Loop style in SET Menu to BOSS, set LOOP to non-quantised in TRK Menu. With the clock running, hit REC, and play in a selection of sounds, such as the notes on a piano one after the other. When finished hit the LOOP button.

Go to SLICE EDIT in the TRK menu, and toggle down to AUTOSLICE, long press the JOG Wheel to start the process. When finished, markers are placed at the transient start points of the all sounds you have recorded. Now you can use a CV to select an individual sound or note within the sample:

In the Modulation menu, find TRIG1 and select the RST option, and remember to toggle back to the TRIG1 option to make the setting active. Then find MOD1 and select SLICE_Q, again toggle the menu back to MOD1 to activate the setting.

In the Track menu set LOOP Mode to 1-SHOT POLY. Connect a CV to MOD1, with the attenuator turn full CW, and connect a GATE to the T1 input. Now you can play the multi-sample.

Alternatively if you have a MIDI keyboard, in the Modulation Menu find MIDI NOTE, and set this to SLICE + RESET. This maps the keyboard note values to the slice values in the multi-sample. Slice 1 is mapped to C2 on the MIDI

keyboard, you can adjust the Slice parameter to offset this value if required.

NB make sure you have set the MIDI CH setting to the correct MIDI Channel for your keyboard or sequencer.

Also to use the MIDI input make sure the Clock Source in the Settings menu is set to Internal.

You can find examples of multi-sample files in the "1-Shot" directory in internal memory such as "1-shot_piano".

Live Granular Buffer [Live Loop]

in the Track Menu menu set the LOOP MODE to LIVE.

Make sure the clock is running.

Set the LOOP length to a short value such as 1/32 or 1/16 etc.

Press the LOOP Button to activate the loop, then press the REC Button to activate recording.

In the Modulation Menu, select TRIG1 and set the destination to RST.

Connect a random or periodic pulse signal to the T1 input of the selected track.

Each time the track receives a high pulse on the T1 input, the Loop will be updated with the incoming audio.

You can even create a live reversed audio effect, by reversing the loop (hold the SHIFT Button and press the MUTE Button)

back up to the top level of the menu to applying the setting.

Connect a randomly varying voltage, or random trigger signal to the T1 input jack, you should start to see the REC LED flashing on and off, indicating the record status. If it is nearly always on, adjust the speed or level of your random source to a slower / lower setting.

Connect a sound source that you can easily manipulate the frequency and timbre, such as the ALGO module, to the track Audio Input, and begin to play with the timbre and frequency controls.

Each time the Record gets triggered little bits of sound begin to get recorded into the loop, because these are quantised in time the sounds will feel quite rhythmic as they build up.

Try occasionally turning off the sound source, to create silent gaps within the loop.

Quantised Record overdubbing to create rhythmic sound patterns

Set a track's LOOP UNIT to QUANTISED, and set the Loop length to 2 bars.

Start the clock, using the S/S button, and press the LOOP button to engage the loop.

In the Modulation menu, navigate to TRIG1, and select REC. Hold the MOD button and using the JOG wheel adjust the quantised parameter to Q1/16. Remember to toggle

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MENU MAP

ALTERNATIVE FUNCTIONS

